

Candidate10 Evaluation Figures

Generated from correctness-gated simulator evidence

Claim rule. Every plotted performance row must pass the architecture-precision oracle, the independent mathematical oracle, and the cross-system final-state check. Compact slices and full datasets are reported as separate populations.

1 Cross-algorithm E2E ranking

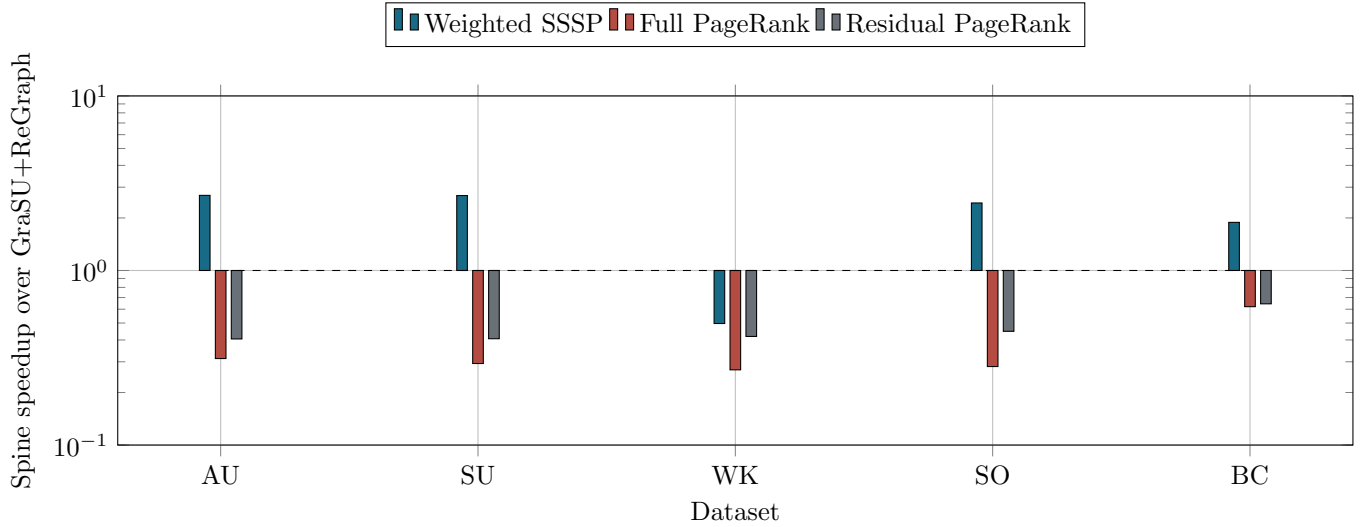
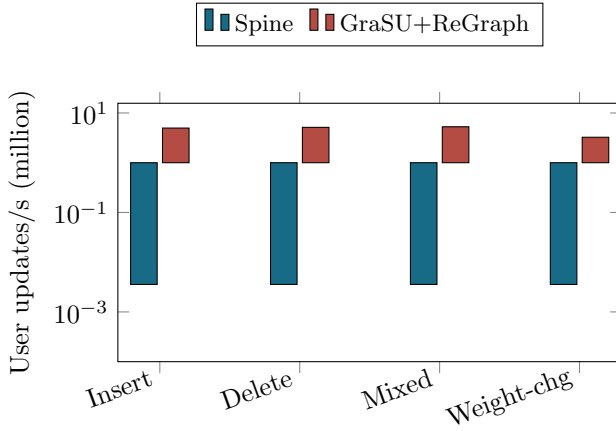


Figure 1: Insert-batch-8 E2E comparison on five compact temporal slices. Values above one favor Spine; values below one favor GraSU+ReGraph.

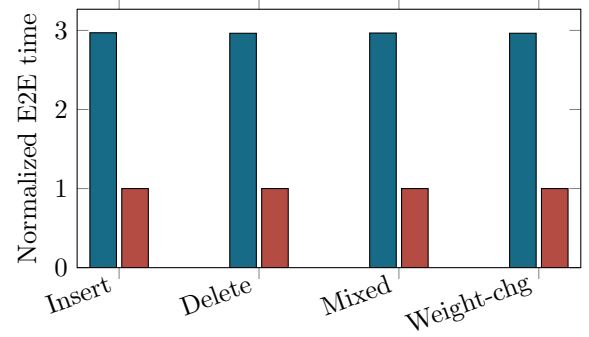
Dataset	Abbrev.	Slice vertices	Base edges	Insert pool
AskUbuntu	AU	3,409	8,192	4,096
Super User	SU	3,224	8,192	4,096
WikiTalk	WK	5,930	8,192	4,096
Stack Overflow	SO	2,004	8,192	4,096
Soc-Bitcoin	BC	10,326	8,192	4,096

Table 1: Compact real-topology slices used by Fig. 1. Each slice contains 8,192 unique non-self base edges plus a separate 4,096-edge later-event insertion pool; the figure uses an eight-edge insertion batch.

2 Update throughput



(a) Semantic update throughput.



(b) E2E time; lower is better.

Figure 2: Full PageRank over four batch-8 differential scenarios and five compact real-topology slices. Every one of the 20 architecture pairs passes both numerical oracles, final graph-state checks, and cross-system rank comparison. Weight change counts eight semantic updates but lowers them to 16 delete-old/insert-new physical records.

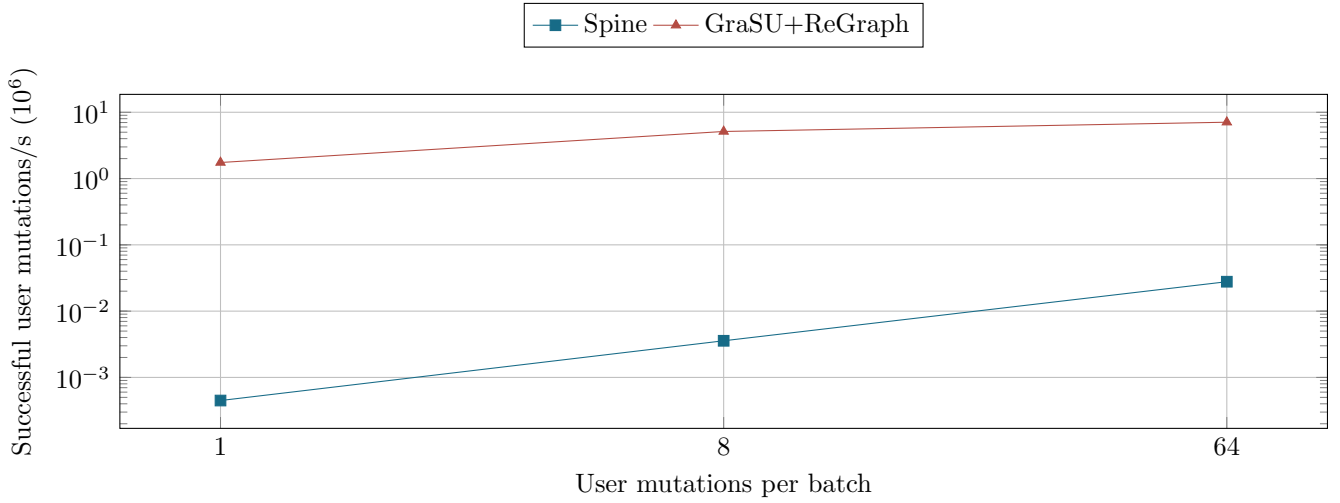


Figure 3: Update throughput versus batch size. The denominator is the number of successful user mutations, not internal physical records.

3 Small-batch scaling

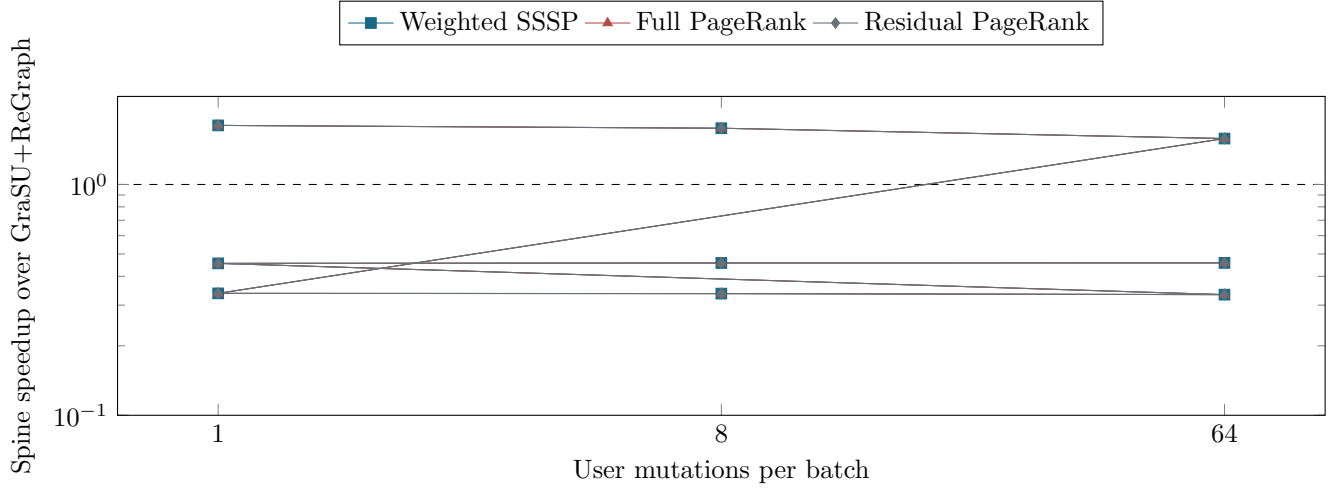


Figure 4: Insertion E2E scaling over five compact real-topology slices. Values above one favor Spine. Each point is the geometric mean of five correctness-gated architecture pairs.

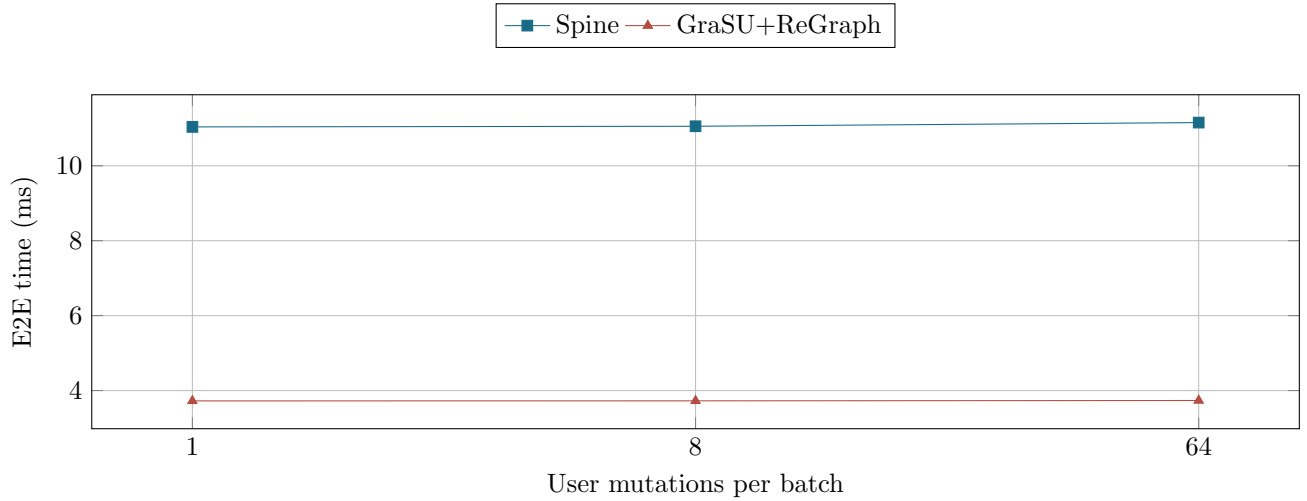


Figure 5: Full PageRank E2E scaling over the five GraSU compact temporal slices. Batch 1 includes insert/delete; batches 8 and 64 additionally include mixed updates.

4 Memory traffic and locality

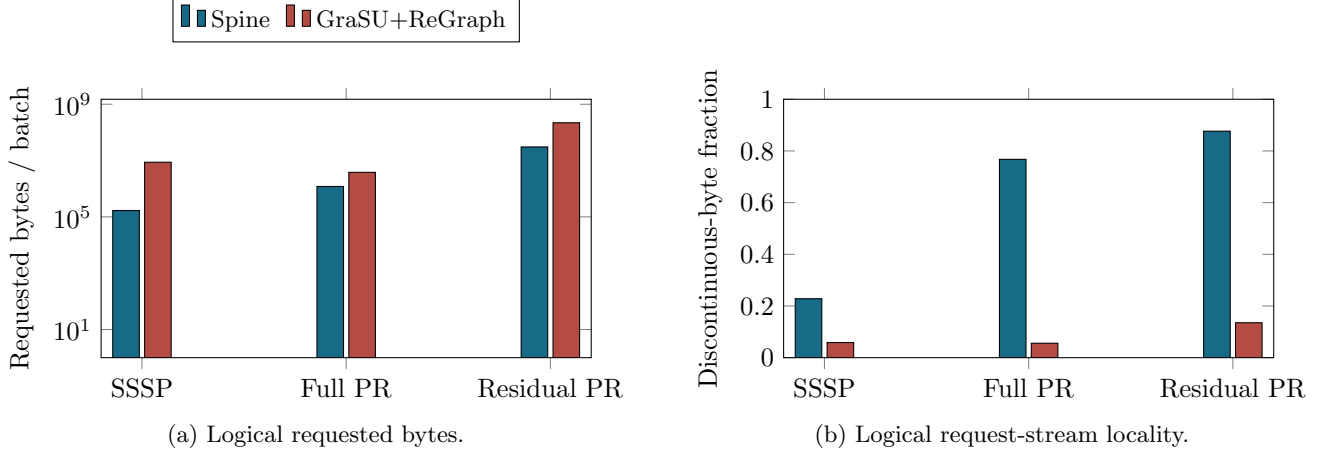


Figure 6: Phase-aligned accepted backend traffic for insert batches of eight over five compact real-topology slices (mean bytes per pair). Locality is classified per initiator and operation after excluding each stream’s first request. It is not a DRAM row-hit metric and excludes burst amplification.

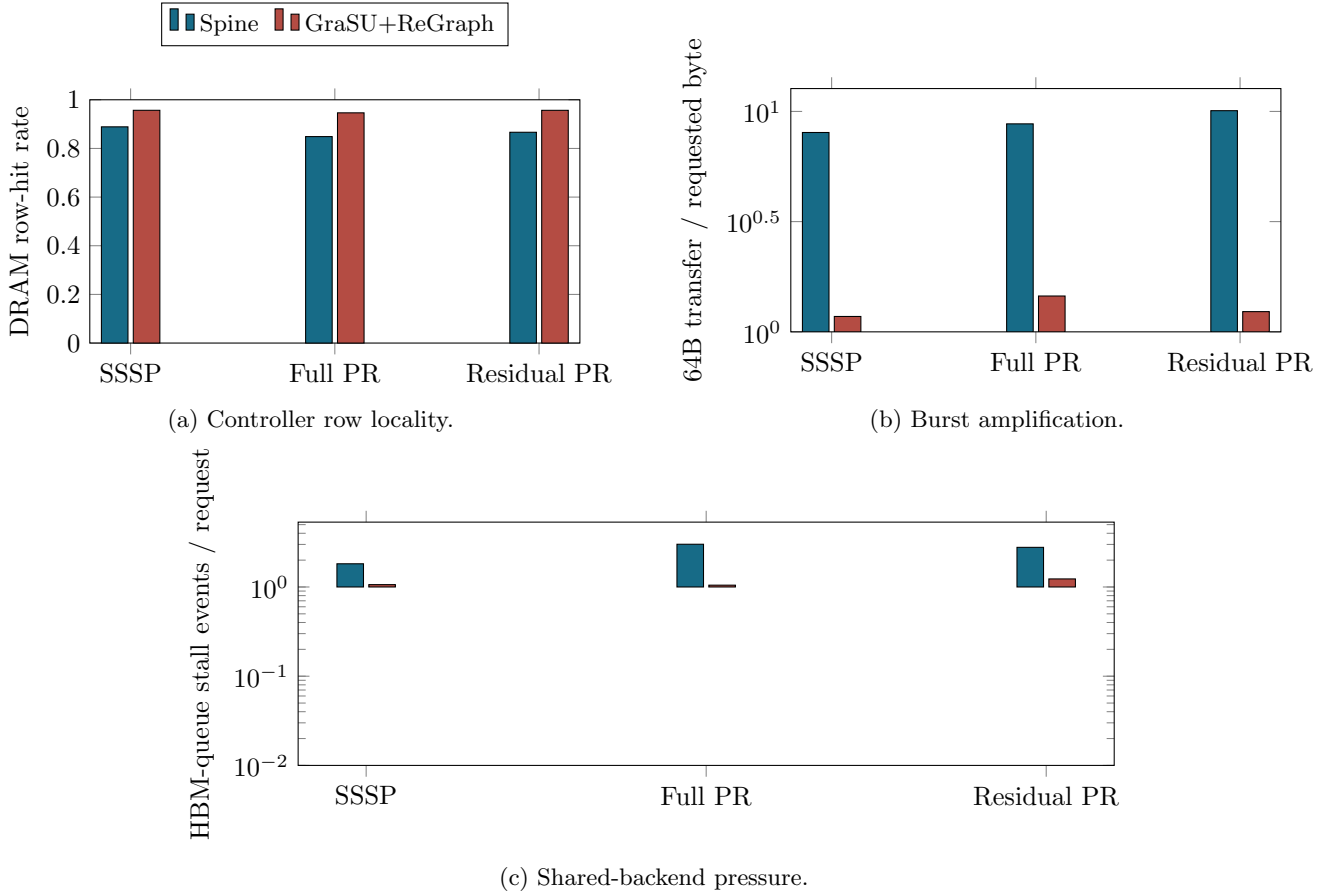


Figure 7: Phase-aligned physical HBM behavior for batch-8 insertion on five compact real-topology slices. The DRAM-Sim3 command ledger equals accepted AXI requests on every row. Stall values count rejected port attempts, so multiple events may occur in one cycle. Spine weighted SSSP uses an exact quiescent cold-prefix subtraction to align its controller window.

5 Dense-batch behavior

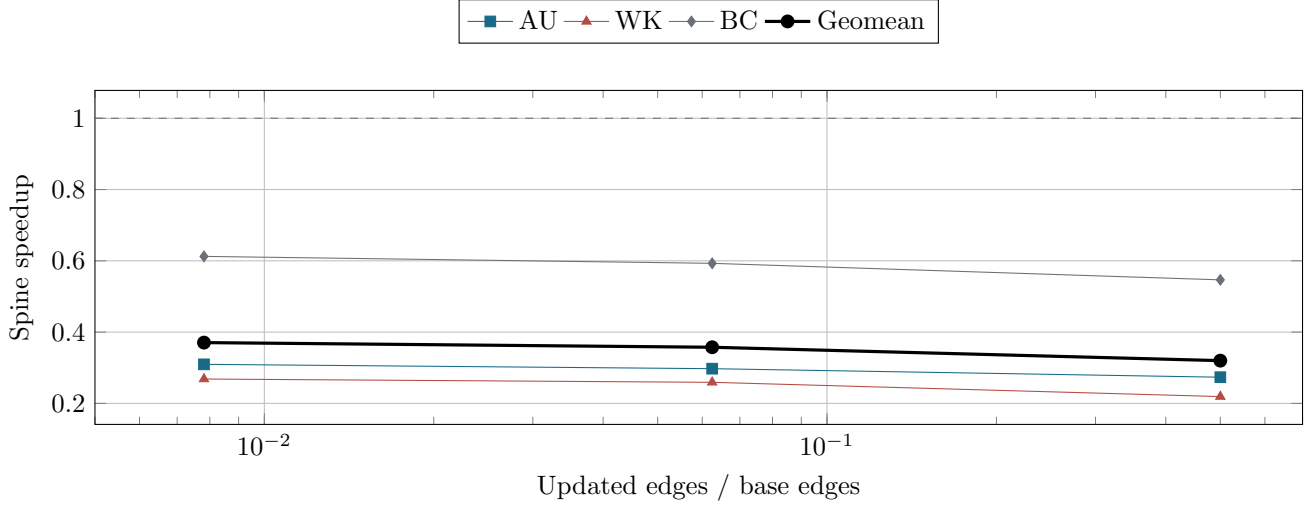


Figure 8: Dense-update crossover on three compact real topologies. Absolute batch size is retained in the companion data table.

6 Matched HBM energy

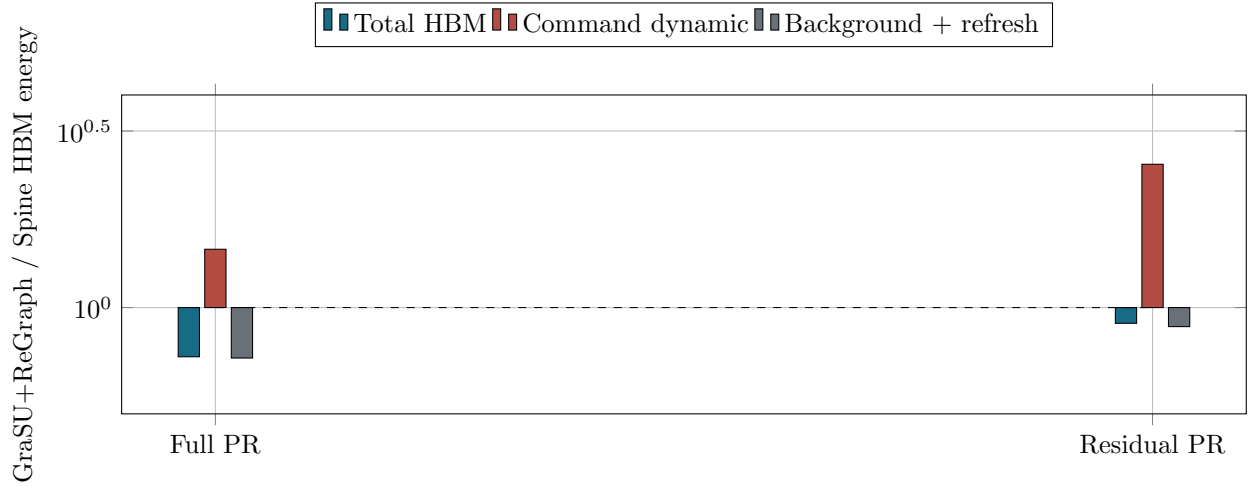


Figure 9: Matched all-32-controller DRAMSim3 energy on three compact holdout slices. Values above one mean GraSU+ReGraph uses more energy. This is HBM energy only, not total accelerator or board energy; asymmetric projected on-chip energy is intentionally excluded.

7 Routed HLS feasibility

Build	Target (MHz)	LUT	REG	BRAM	URAM	DSP	WNS (ns)	Setup
Spine SSSP	152	127,847	149,744	99	99	22	0.003	closed
G+R SSSP	150	96,559	107,380	233	64	0	-0.013	target missed
G+R Full PR	150	178,374	177,836	278	64	304	-0.13	target missed
G+R Residual PR	150	247,692	301,029	293	64	336	-0.087	target missed

Table 2: Routed U55C HLS feasibility at each row’s stated data-clock target. Spine is the latest source-identical 152 MHz implementation; the GraSU+ReGraph implementations target 150 MHz. These builds are not iso-functional: Spine is the native SSSP core baseline, whereas the GraSU+ReGraph rows are algorithm-specific normalized whole systems. Absolute resource ratios are therefore not used as a comparative performance claim.

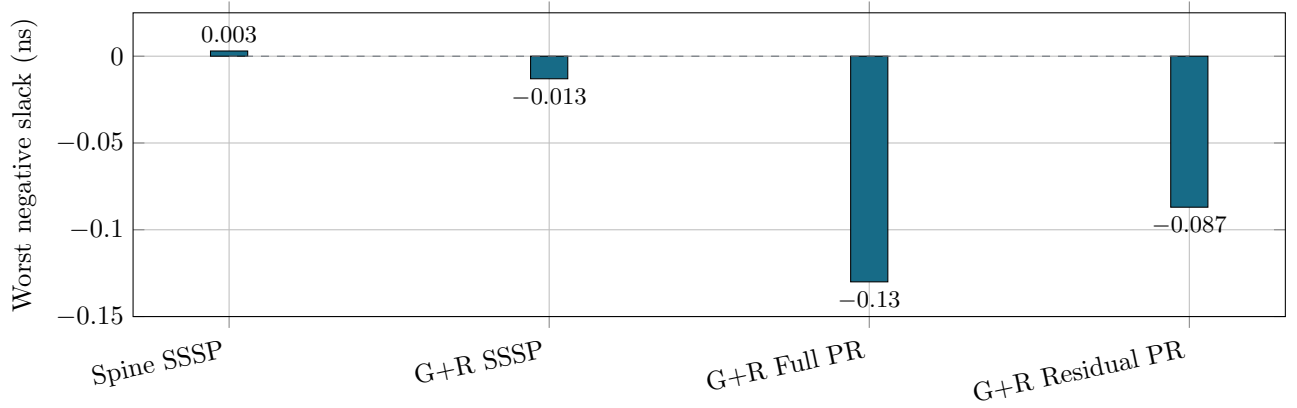


Figure 10: Post-route setup margin at each row’s stated target (152 MHz for Spine and 150 MHz for GraSU+ReGraph). Non-negative WNS closes timing; negative WNS identifies the remaining target miss.